

ABDUL RAZAK

Staff Data Scientist, Extreme Networks

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Portfolio | Google Scholar | DBLP | ORCID | Patents | LinkedIn

RESEARCH SUMMARY

Data scientist with more than a decade of experience taking machine-learning systems from prototype to production, backed by a research career in decision making under uncertainty. At Extreme Networks I lead the data-science work on generative models for network diagnosis. At Johnson Controls I built the probabilistic and rule-based models behind eight granted US patents in building security. The algorithms for multi-objective optimisation and preference inference that I developed during my doctorate and post-doctoral fellowship, published at AAIL, IJCAI, UAI, CP and ICTAI with colleagues at IBM Research Dublin, still shape how I model trade-offs in production.

Research interests. Multi-objective decision making under uncertainty; preference elicitation and inference; probabilistic graphical models; machine learning; large language models and generative AI.

EDUCATION

- 2010–2014 **PhD, Computer Science (Artificial Intelligence)**
National University of Ireland, Cork. Cork Constraint Computation Centre
Thesis: *Reasoning with Imprecise Trade-offs in Decision Making under Certainty and Uncertainty*, supervised by Nic Wilson. Funded by the IRCSET / IBM Enterprise Partnership Scheme.
- 2003–2005 **MSc, Mathematics with Computer Science**
Osmania University, Hyderabad, India
Awarded the university merit scholarship.
- 2000–2003 **BSc, Mathematics, Statistics and Computer Science**
Osmania University, Hyderabad, India

PROFESSIONAL APPOINTMENTS

- Since 2020 **Staff Data Scientist**
Extreme Networks. Scottsdale, Arizona; previously Cork, Ireland
- Lead the data-science work on generative models for network diagnosis. Language models generate and explain hypotheses; anomaly detection, a knowledge graph and live telemetry supply the evidence they reason over.
 - Built agent workflows that triage client connectivity, roaming and device-health problems across thousands of access points and switches, and draft remediation steps for network operators.
 - Developed correlation and inference pipelines over telemetry and client-experience metrics, so that investigation of routine faults takes seconds rather than hours.
 - Adapted the preference-inference methods from my research to rank recommendations for customers whose priorities differ in reliability, cost and coverage.
- 2017–2020 **Data Scientist**
Johnson Controls. Cork, Ireland
- Machine-learning lead for the intrusion-alarm, access-control and retail-analytics product lines.
 - Alarm-panel classification by Pareto optimisation, root-cause analysis of events, and site-independent signature generation. Eight granted US patents issued from this work.
 - Relational probabilistic models in BLOG and Bayesian inference in PyMC3, deployed across large multi-site estates.
- 2014–2017 **Post-Doctoral Fellow, Science Foundation Ireland**
Insight Centre for Data Analytics, NUI Cork
- Probabilistic graphical models, clustering and influence diagrams for industrial decision problems, within a programme run with United Technology Research Centre.
 - Preference inference with lexicographic and weighted-average models, partly with IBM Research Dublin.
- 2005–2010 **Lecturer in Mathematics**
Osmania University. Hyderabad, India
- Undergraduate teaching in algebra, real and numerical analysis, calculus, probability and statistics.

PEER-REVIEWED PUBLICATIONS

Listed most recent first. Complete record on [Google Scholar](#) and [DBLP](#).

- 2017 R. Marinescu, **A. Razak**, N. Wilson. Multi-objective influence diagrams with possibly optimal policies. *Proceedings of the 31st AAAI Conference on Artificial Intelligence*.
- 2015 Anne-Marie, **A. Razak**, N. Wilson. The comparison of multi-objective preference inference based on lexicographic and weighted average models. *Proceedings of the 27th IEEE International Conference on Tools with Artificial Intelligence (ICTAI)*.
- 2015 N. Wilson, **A. Razak**, R. Marinescu. Computing possibly optimal solutions for multi-objective constraint optimisation with trade-offs. *Proceedings of the 24th International Joint Conference on Artificial Intelligence (IJCAI)*.
- 2015 Anne-Marie, **A. Razak**, N. Wilson. Multi-objective constraint optimisation with lexicographic preference models. *Workshop on Algorithmic Issues for Inference in Graphical Models (AIGM)*.
- 2013 R. Marinescu, **A. Razak**, N. Wilson. Multi-objective constraint optimisation with trade-offs. *Proceedings of the 19th International Conference on Principles and Practice of Constraint Programming (CP)*.
- 2012 R. Marinescu, **A. Razak**, N. Wilson. Multi-objective influence diagrams with trade-offs. *Proceedings of the 28th Conference on Uncertainty in Artificial Intelligence (UAI)*.

GRANTED US PATENTS

Nine granted between 2019 and 2021, grouped by subject. Filings are searchable on [Justia](#) and [Google Patents](#).

Decision intelligence and maintenance

- 2019 **A. Razak**, L. A. Deleris, R. Marinescu, N. Wilson. Interactive method to reduce the amount of trade-off information required from decision-maker in multi-attribute decision-making under uncertainty.
- 2019 **A. Razak**, G. Subramanian, C. J. Donovan. Virtual maintenance manager.
- 2021 D. Horgan, **A. Razak**, G. Bernardi. Root-cause analysis of event occurrences.

Alarm management

- 2020 **A. Razak**, M. Stewart, G. Subramanian. Automated alarm panel classification using Pareto optimization.
- 2021 **A. Razak**. Systems and methods for managing alarm data of multiple locations.
- 2021 M. Costello, F. Fala, J. R. Holliday, N. B. Pavel, **A. Razak**. Building security system with alarm lists and alarm rule editing.

Building security analytics

- 2020 M. Dziduch, C. J. Donovan, **A. Razak**. Building security analysis system with site-independent signature generation for predictive security analysis.
- 2020 M. Dziduch, C. J. Donovan, **A. Razak**. Rules-based method of identifying misuse of emergency fire exits using data generated by a security alarm system.
- 2020 M. Dziduch, C. J. Donovan, **A. Razak**. Systems and methods for managing a security system.

HONOURS AND FUNDING

- IEEE Senior Member, elevated May 2026.
- Science Foundation Ireland post-doctoral fellowship, Insight Centre for Data Analytics, NUI Cork, 2014 to 2017.
- IRCSET / IBM Enterprise Partnership Scheme scholarship, full doctoral funding, 2010 to 2013.
- Osmania University merit scholarship (MSc, 2003 to 2005), and 40th place of roughly 30,000 candidates in the university mathematics entrance examination.

ACADEMIC SERVICE AND INVITED TALKS

- Reviewer, *Computers and Electrical Engineering* (Elsevier). More than ten manuscripts in artificial intelligence, wireless networking and intelligent systems.
- Reviewer, IEEE International Conference on Communications (ICC), artificial-intelligence submissions to the 2025 technical programme.
- Invited talks: Sapienza University of Rome, on preference handling in decision making under certainty and uncertainty; NUI Galway, on probabilistic decision making under uncertainty; NUI Cork and IBM Research Dublin, during the Risk Management Collaboratory and United Technology Research Centre programmes.
- Conference presentations, oral and poster, at AAAI, IJCAI, UAI, CP and ICTAI.
- Five years of undergraduate lecturing, and mentoring of junior data scientists in industry.

TECHNICAL SKILLS

Programming	Python, C++ (STL, Boost, Eigen), SQL, PySpark, PyFlink.
Modelling	PyMC3, BLOG, Swift, ILOG CPLEX, LPSOLVE, scikit-learn.
Platforms	AWS SageMaker and Athena, Azure Databricks, streaming and batch pipelines.
Languages	English, Hindi, Telugu, Urdu. Irish citizen since 2021.